

COMPUTER ORGANISATION AND **ARCHITECTURE ASSIGNMENT (12 QUESTIONS):**

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Q1:What is Raspberry pi?

Raspberry Pi is a low-cost mini computer that makes it easier for individuals to learn programming and electronics. Despite its small size, Raspberry Pi can perform a variety of functions such as browsing the internet, programming, robotics, artificial intelligence development, and internet of things amongst others. Individuals appreciate using this computer because of its ease to use and low power consumption.

Q2. What are the 10 takeaways of the AI Summit?

AI (Artificial Intelligence) is going to be applied in real-world situations.

Creation of AI should be safe and responsible.

AI is seen as investment for economic growth.

Creation of computing infrastructure to support AI is critical.

Skills related to AI are important for career development.

AI contributes to increased productivity and eliminates routine tasks.

Industry, Government and Researchers should collaborate on AI.

Security, privacy and transparency issues on AI should be addressed.

There is a need to invest in continuous research and development of AI.

AI will have a significant impact on Technology, Business and Society

Q3. How do you choose a programming language to design an application, and what are the important parameters?

To choose a programming language one should consider the following parameters:

- Application field or area
- Speed and performance
- Secrecy and reliability

- Operating system and platform
- Readability or writability
- Existence of libraries, tools and documentation
- Support and Maintenance
- Cost of development and maintenance

For instance, C programming language is used in computer hardware, Java is used in desktop and mobile application and Python is used in AI. All this makes sense because choosing the best programming language is key to developing applications effectively and efficiently.

Q4. What is NPTEL?

NPTEL (National Program on Technology Enhanced Learning) is an online education portal initiated by IITs and IISc Bangalore under the funding of the government of India, which provides online courses to learners all over the country. These courses are offered in various streams such as engineering, science, management, humanities, and social sciences at no cost. Students learn using videos, assignments, and course material in their own time and convenience. There is also an option to take an optional exam to receive a certificate of completion of the selected course.

Q5. What is the meaning of RAM size, Word size, if the system is of 64bit, what is the speed and operating system of our PC?

RAM Size: This means the amount of Random Access Memory that is present in a computer or mobile device.

Word Size: This refers to the amount of information that the CPU can handle at one time. If our system is of 64bit, then it means it can handle 64 bit data at a time and this makes it fast.

Speed: The speed of a computer is mainly determined by the CPU. The higher the speed of a computer is, the more powerful it is compared to another one with a lower speed.

Operating System (OS): This is the interface between the computer user and the hardware. It manages processes, memory, CPU, and other peripherals. For instance, the known operating systems include Windows, macOS, and Linux amongst others.

Q6.What is the Stampeding Herd Problem?

The Stampeding Herd Problem is a scenario whereby a number of threads and processes try to access the same resource. Once that resource becomes available again, all the threads and processes that were waiting for it will access it at the same time. All run concurrently and this increases CPU utilization, which may slow down the system performance. I envision this problem in real-life situations in cases where students are outside a classroom waiting for the teacher to open the door. The moment the door is opened all of them rush and try to get in at the same time. This creates congestion and delays since they are all trying to enter at the same time. Similarly, this analogy applies to the stampeding herd problem in a computer system. This happens when there are a number of processes competing for one resource at a time

Q7. What is the Turing Award? Who received it recently?

The A.M. Turing Award is one of the most prestigious awards in computer science; it recognizes outstanding contributions to the field of computer science. It was established by the Association for Computing Machinery (ACM). According to the assignment, Andrew G. Barto and Richard S. Sutton received the Turing Award recently for their work in Reinforcement Learning.

Q8. Who are the founders of Intel and AMD?

Intel was founded in 1968 by Gordon Moore and Robert Noyce. Meanwhile, AMD was founded in 1969 by Jerry Sanders and seven other co-founders.

Q9. CPU vs GPU vs TPU

ANS: The CPU is used for computer operations such as running applications and operating systems. The GPU is a processor with cores that handles graphic and heavy operations. The TPU is a specialized processor made for AI workloads. The CPU is versatile, the GPU is good for graphics, and TPU is efficient in AI processing.

Q10. The Imitation Game

ANS: After watching The Imitation Game I realized that to be successful one needs to be intelligent, persistent and patient. The movie highlights how Alan Turing overcame all odds to achieve his goals simply because he had the determination and the knowledge. His breakthrough in history was made possible by his hard work and brilliance. In addition, I learned the importance of teamwork in day to day life. Though Alan Turing was highly intelligent, he would not have achieved his goals if teamwork was not emphasized. The movie made me realize that hard work and teamwork go hand in hand.

Q11. What is Yotta?

ANS: Yotta is a company that provides cloud and data center services. It has a number of unique services offered which include:

- Cloud computing services
- Data security
- Energy efficient data centers
- Disaster recovery options

Yotta is deployed in various fields such as Artificial Intelligence, healthcare, finance, government, public sectors, education and research.

Q12. Why is GPU faster than CPU?

ANS: The CPU is the mastermind of the computer and it deals with all other processes. The GPU is a graphics processor which is used to carry out operations faster and it has cores for faster processing of heavy tasks. This is why the GPU is faster than the CPU.